Machine learning project

Predicting the french electricity consumption from the weather analysis

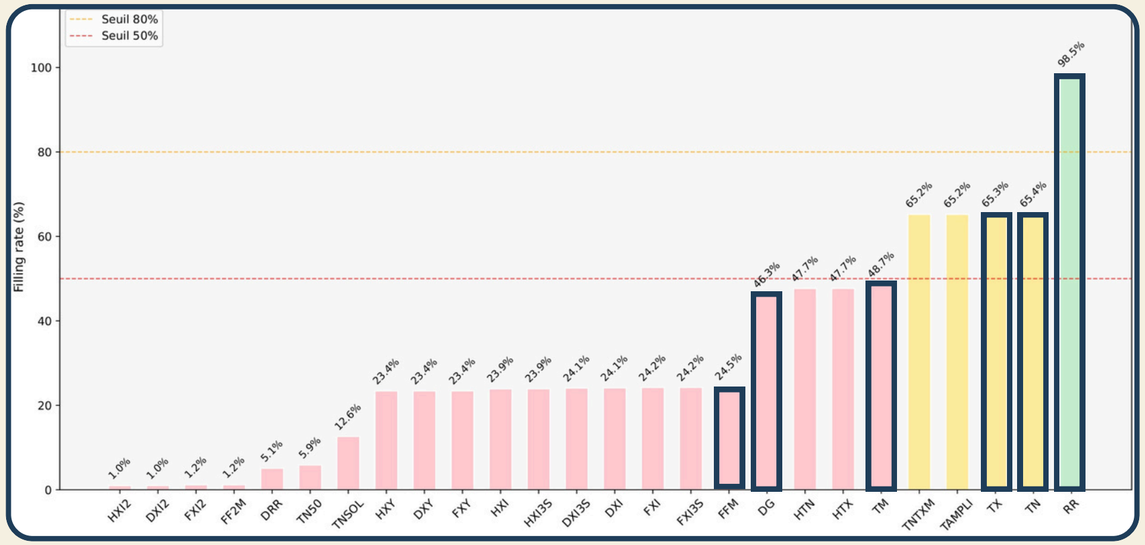
Félix de Batz

April 2026

Converting raw weather data

Daily

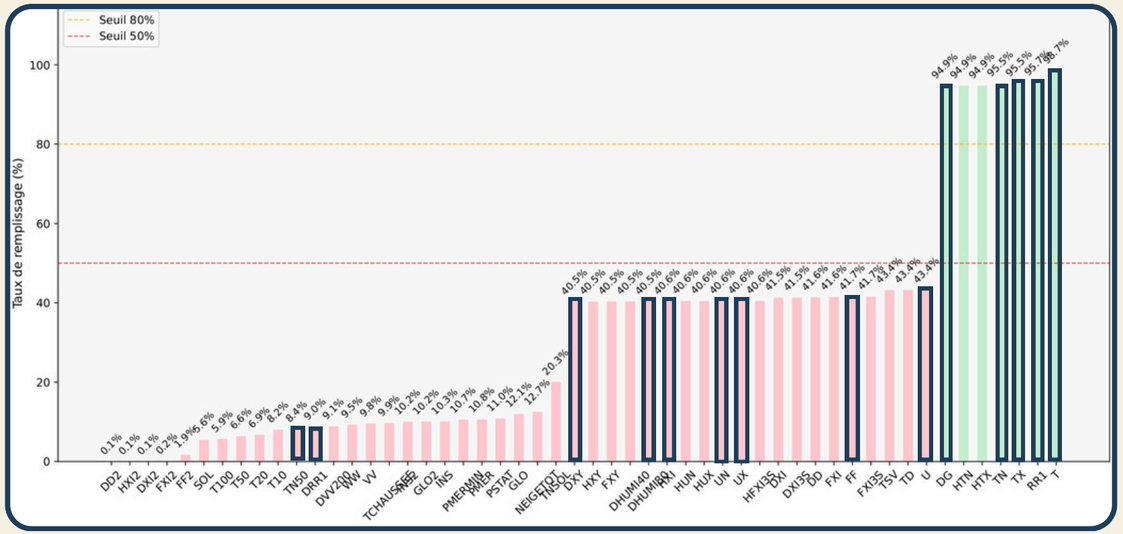
2013 - 2024



7 columns - 4383 rows

Hourly

2020 - 2024



7 columns - 43848 rows

Creating datsets

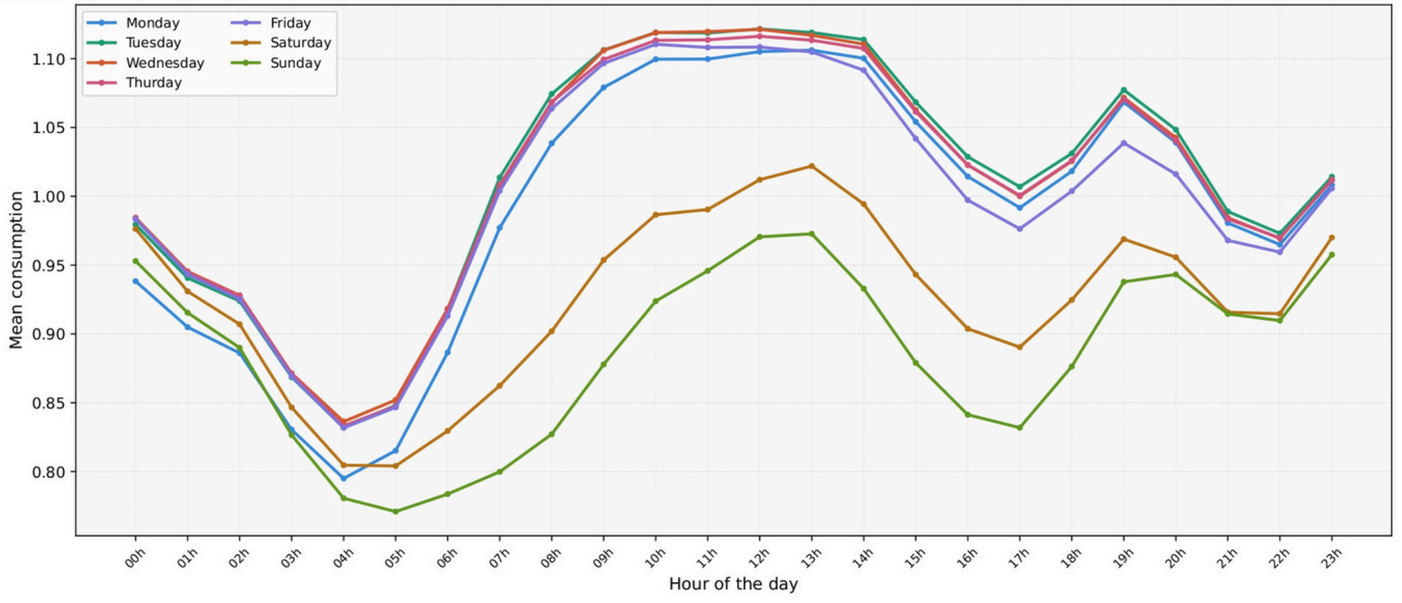
Converting then merging the energy and weather datasets

4 datasets per region

Mean scale \ Timming scope	Daily	Hourly
Region	7 x 4383	16 x 43848
Department	(26-80) x 4383	(62-158) x 43848

For the big datasets we had to introduce a PCA

Important analysis

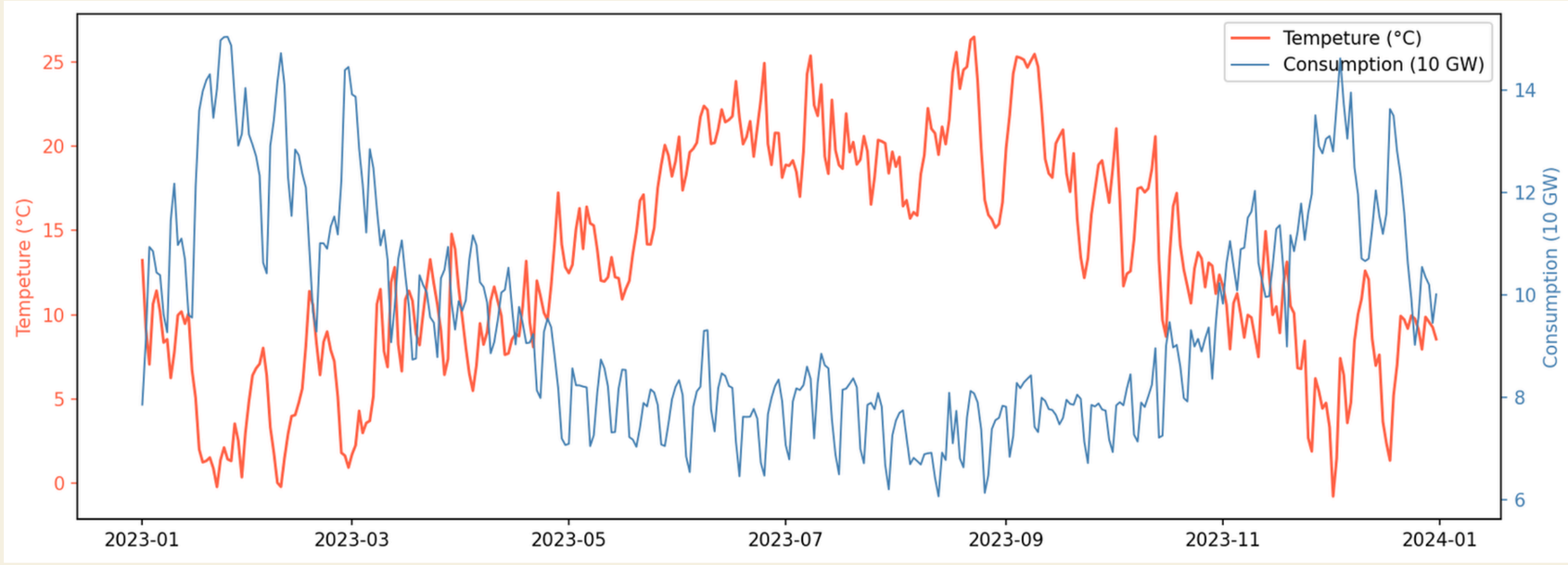


Nouvelle Aquitaine mean consumption by Hour (2022)

Weekdays and hours are important!

New features

Temperature and consumption are correlated



Centre Val de Loire (2023)

The approach

Model used

Polynomial regression

Ridge regression

Regression Trees

AdaBoosting on regression Trees

Fine tuning

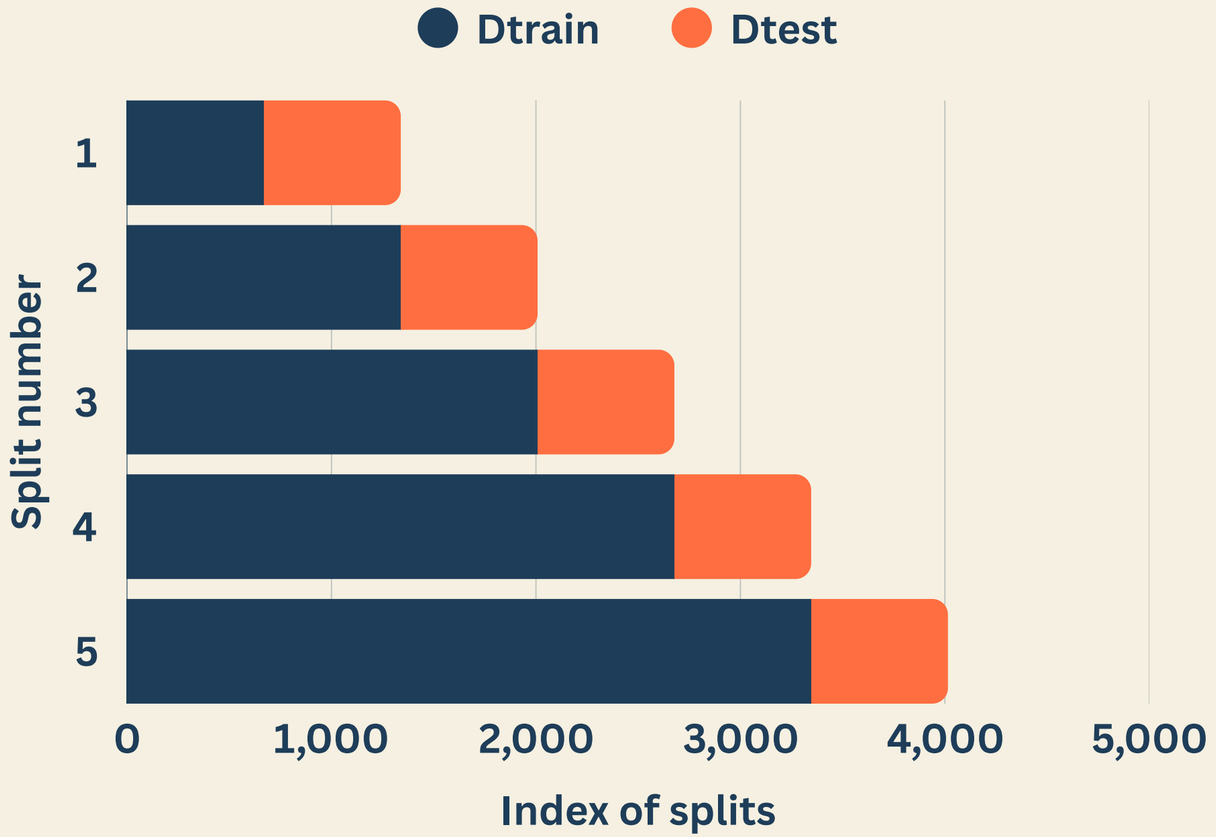
Dtrain : <2024

Test model on each dataset

Cross-validation to find hyperparameters on Dtrain

Testing on 2024

Time series cross validation



Times series example with 5 split on a daily dataset

The results

General takeaways

Good results across the board

More features + PCA worse than simple features

knowing **last consumption** had a big impact

Some regions are **harder** than others

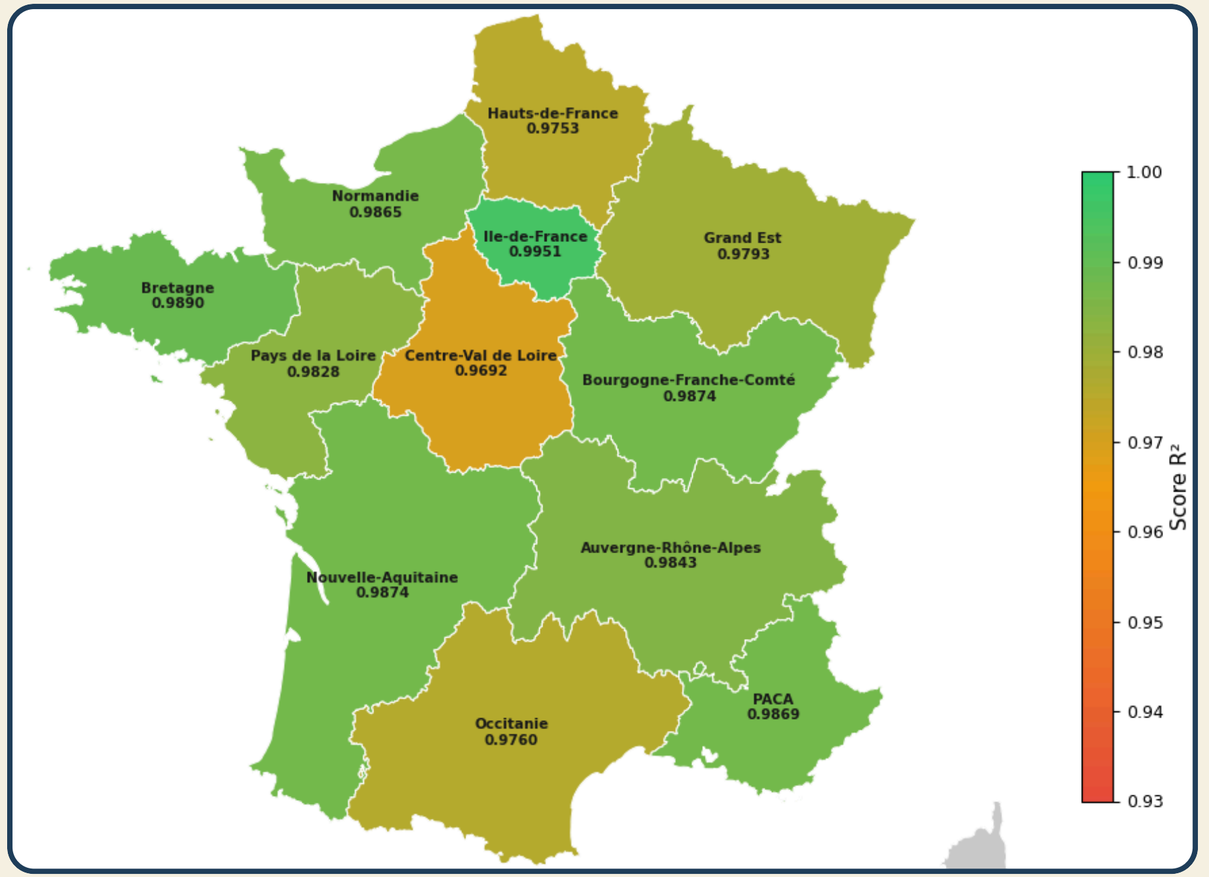
The results

Polynomial regression

Quite good

Pretty fast

Quickly overfit
with high degree



Polynomial results on Region - Hour

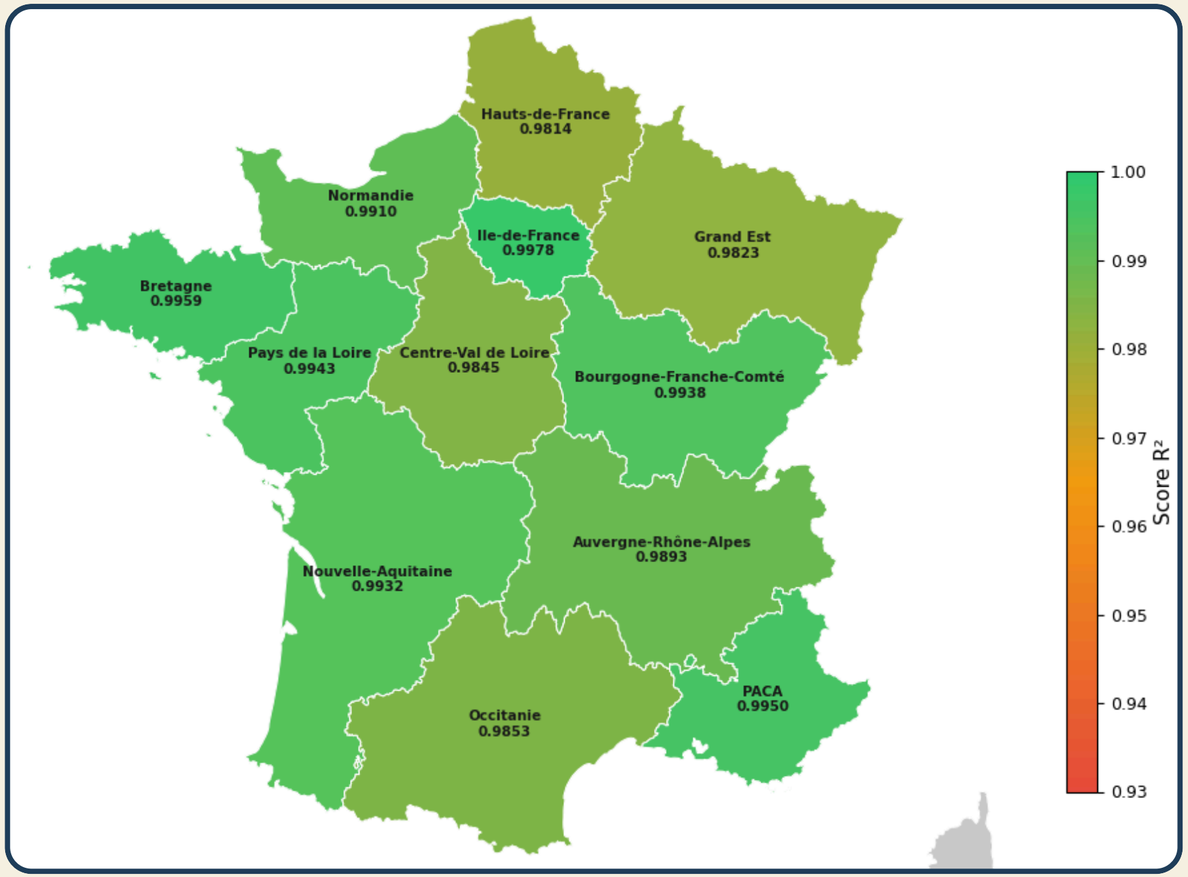
The results

Ridge regression

Really good

Correct the polynomial's errors

Pretty fast



Ridge results on Region - Hour

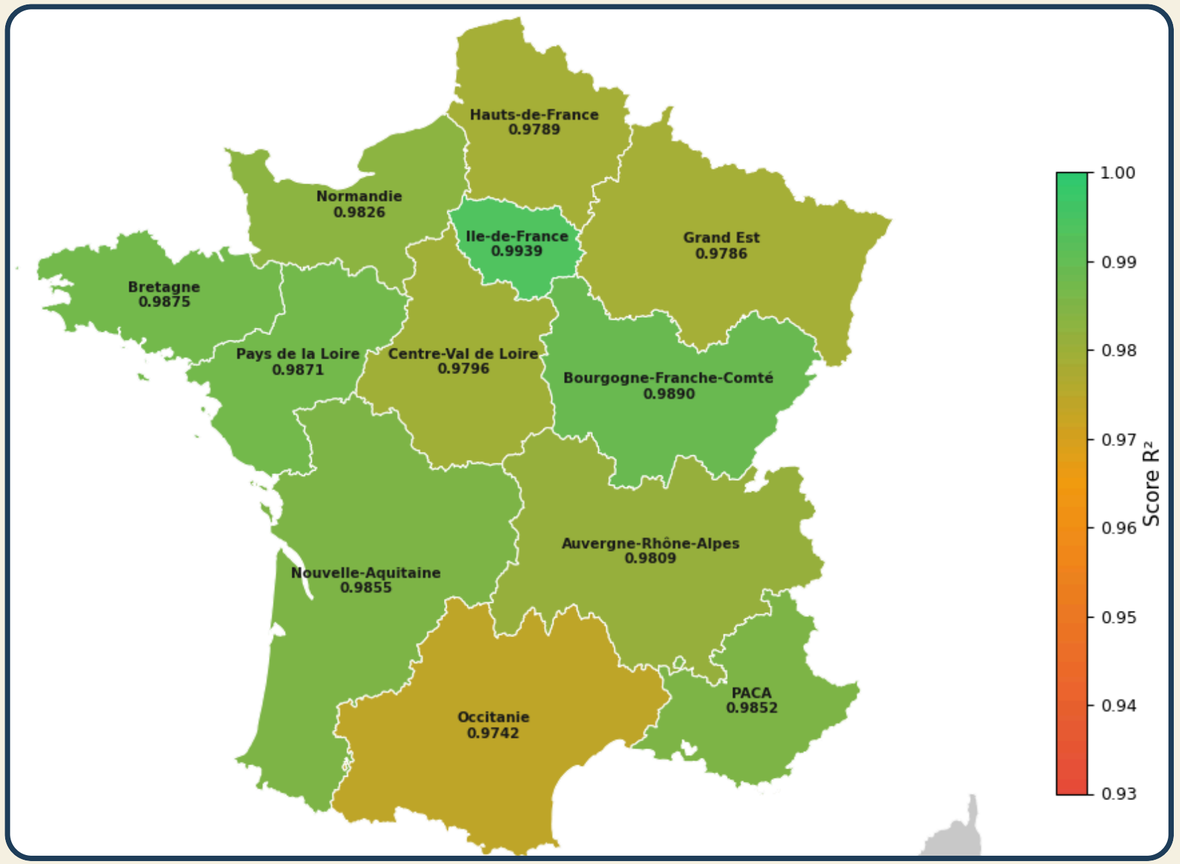
The results

Regression Trees

Not as good

Tends to overfit with high depth

Extremely fast



Regression trees results on Region - Hour

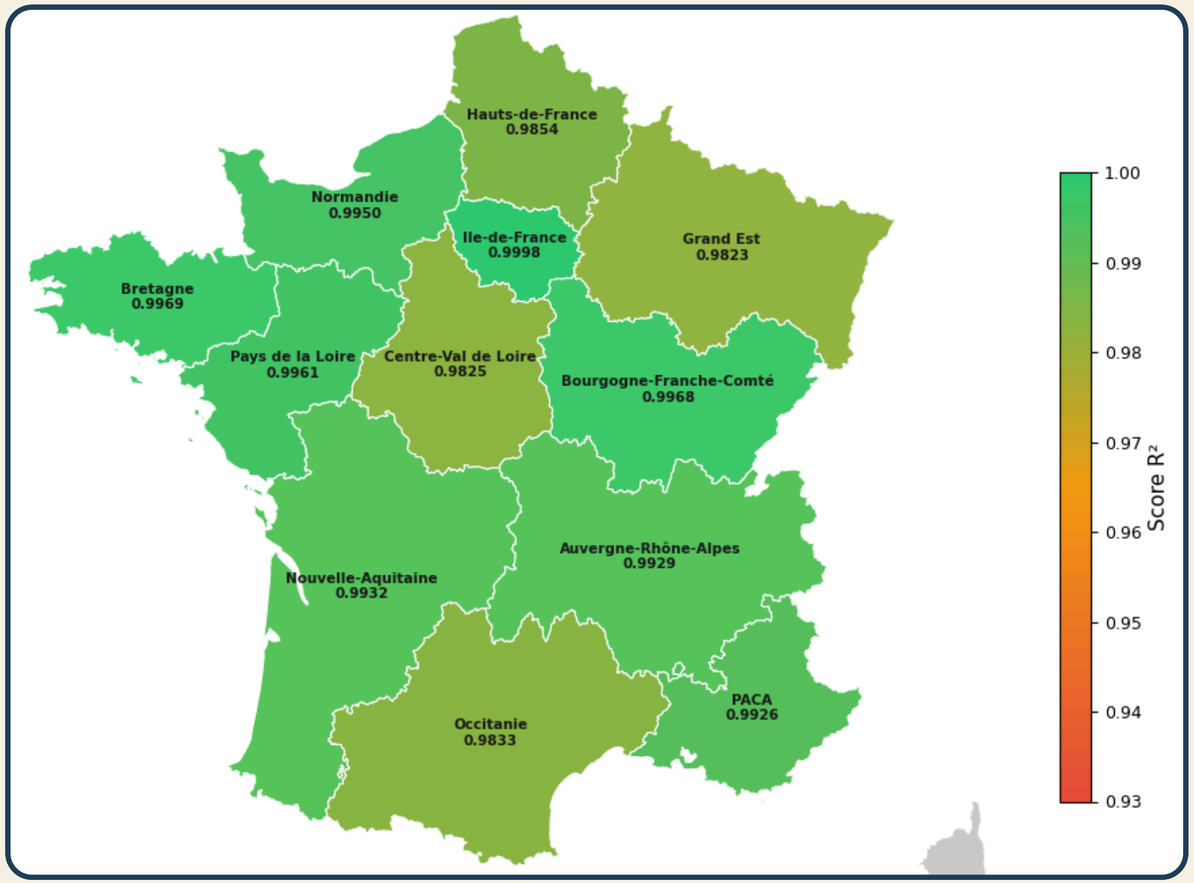
The results

AdaBoosting

Really good

Corrected the trees's errors

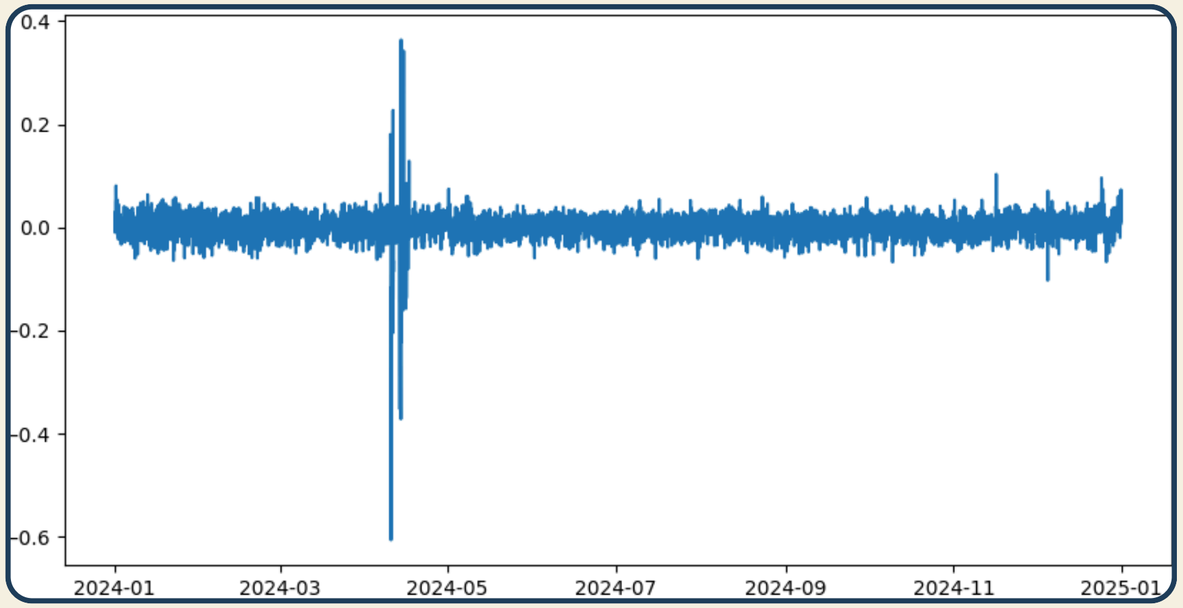
Slow



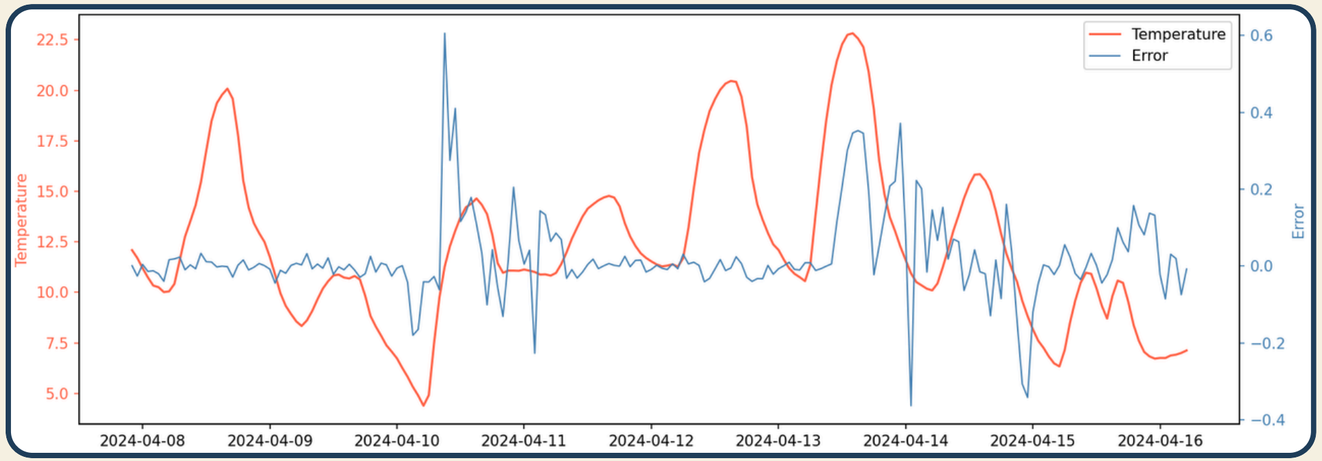
AdaBoosting results on Region - Hour

The results

Analyzing Haut de France Ridge result



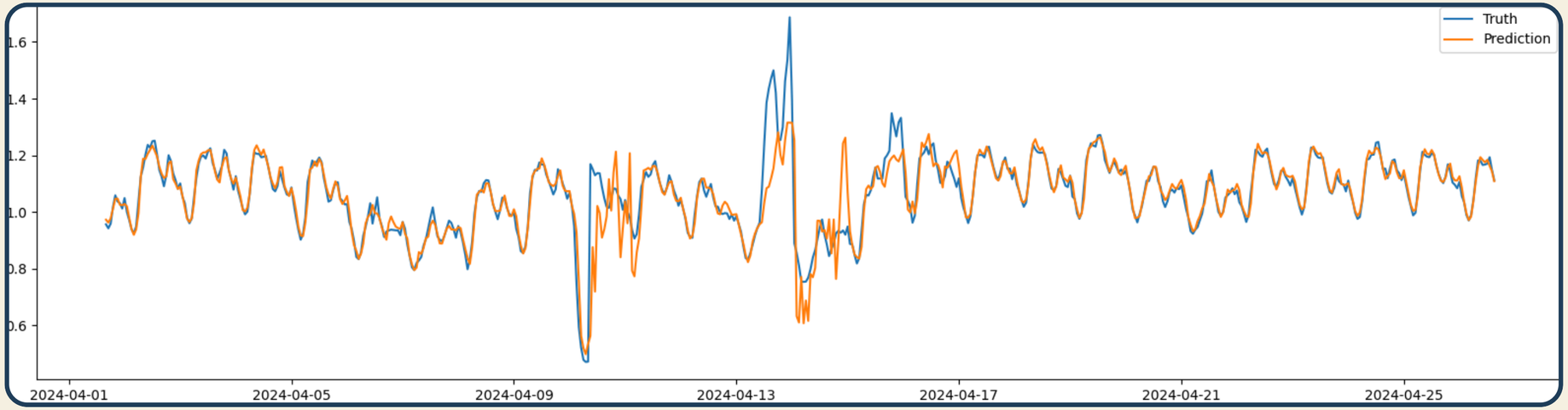
Error curve in 2024



Temperature vs error (zoom)

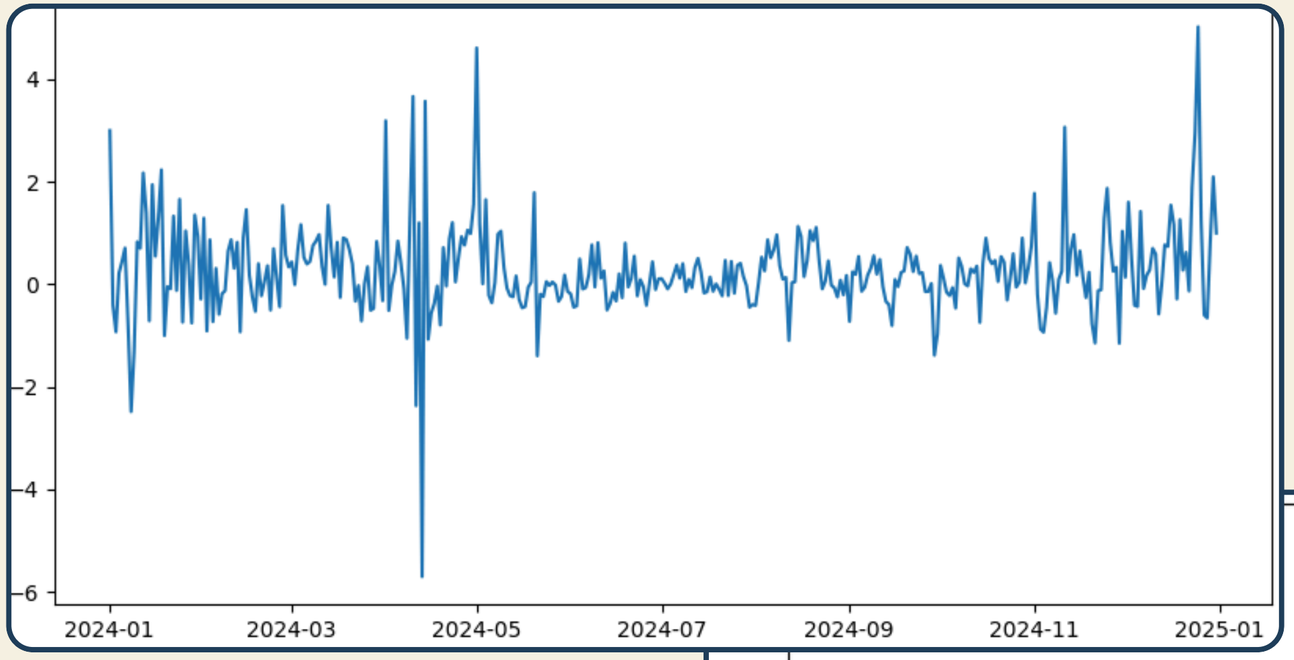
2024 score: 0.981

Prediction vs Truth (zoom)



The results

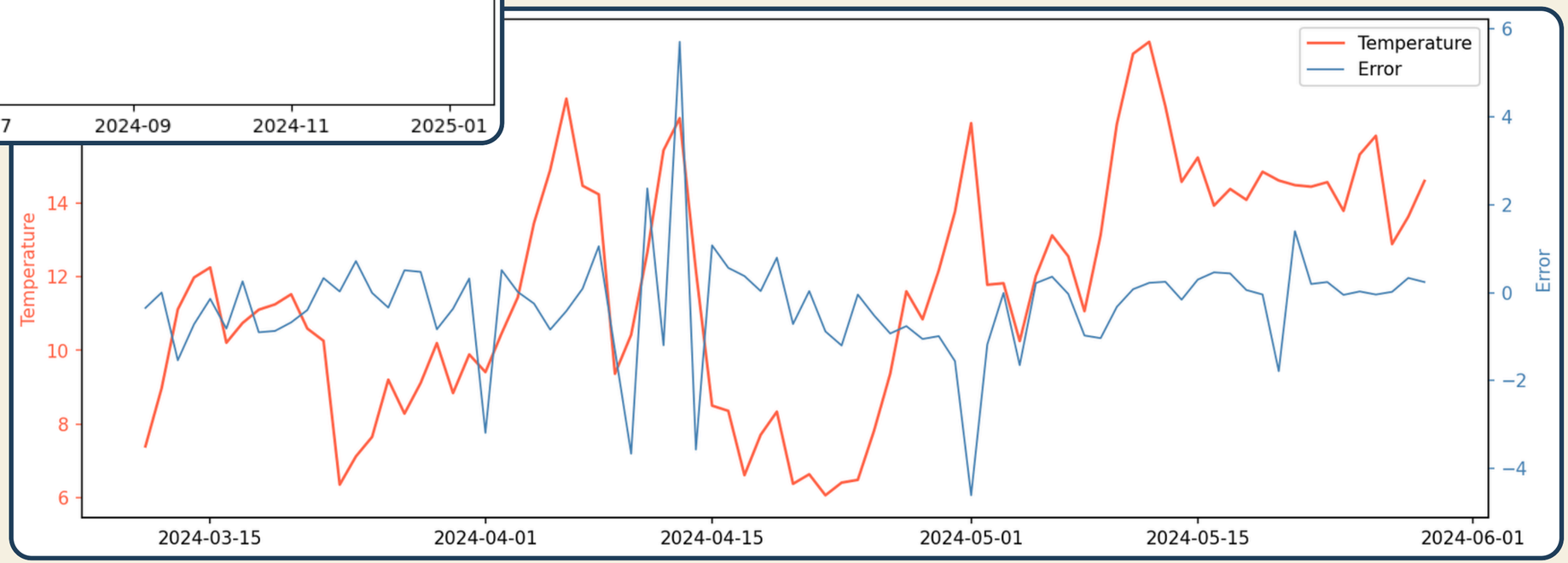
Analyzing Haut de France AdaBoosting result



2024 score: 0.941

zoom score: 0.673

Error curve in 2024



Temperature vs error (zoom)

Side tests

General takeaways

Interpolate missing data by taking the 5 nearest neighbors

More features + PCA worse than simple features

PCA reduction

Some regions are harder than others